DO NOW — RECALL

Lesson 139 · SQL INSERT, UPDATE and DELETE Safely

Name: _____ Class: _____ Date: _____

1. What is the difference between WHERE and ORDER BY in a query?

2. What are the outputs of a half adder when A=1 and B=1?

3. In Big O notation, what is binary search worst-case complexity?

4. When does a linear search stop?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

WHERE filters rows; ORDER BY arranges returned rows.

2. Model answer

Sum 0, carry 1.

3. Model answer

O(log n) for sorted random-access data.

4. Model answer

On a match or after examining all items.

By the end, you can…

• INSERT INTO Table(columns) VALUES(...).

• UPDATE Table SET column=value WHERE key=...

• DELETE FROM Table WHERE key=...; verify matches before writing.

Vocabulary — know these terms

INSERT — Adds new rows.

UPDATE — Changes values in existing rows.

DELETE — Removes matching rows.

WHERE — Restricts affected rows; omission can affect every row.

NEW LEARNING

Read, annotate and explain

Create rows

INSERT INTO Table(columns) VALUES(...).

Change rows safely

UPDATE Table SET column=value WHERE key=...

Delete deliberately

DELETE FROM Table WHERE key=...; verify matches before writing.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A Book table has BookID=8,Stock=4. Write an UPDATE to set Stock=3 for that row.

Identify BookID 8

Find the given data and name the required result.

UPDATE Stock = 3

Explain why this intermediate stage is necessary.

WHERE BookID = 8

UPDATE Book SET Stock = 3 WHERE BookID = 8; Before running, SELECT the matching row. Without WHERE, the stock column of all books would be changed.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write SQL inserting StudentID 22 and Name 'Kai' into Student. [4]

Question 2

A teacher must change a single pupil's YearGroup to 13. Explain a safe SQL statement. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain risks in DELETE FROM Booking; and suggest protections in a relational production database. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

UPDATE Book SET Stock=0; changes exactly one Book record.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.